

# AI Audit Report — HW06

**Student ID:** 23127152

**Declaration:** I use AI tools for the following tasks.

**Công cụ AI:** Cursor (Cursor Grok 4.5)

**Skill đã dùng:** hw06-api-testing, api-test-generate, api-test-audit, api-test-extend, api-test-execute, bug-report, ai-audit-report

**Quy tắc audit:** mỗi tương tác AI được ghi **ngay sau** khi xảy ra.

**Bắt đầu audit:** 2026-08-19

## Tools Declared

| Tool             | Version / Model          | Purpose                                        |
|------------------|--------------------------|------------------------------------------------|
| Cursor           | Cursor Grok 4.5          | Generate / audit assist / Postman suite / bugs |
| Postman + Newman | Newman 6.2.2 · htmlextra | Execute API tests                              |

## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.1 Step A Domain

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:38 +07

**Prompt:**

Using skill api-test-generate Step A only: for FR-05 GET /api/products optional query search, generate domain partition test cases (valid/invalid/boundary/charset/encoding). Spec excerpt from api\_specification.md §3.1. Output rows for generated.md template. Do not generate security or schema yet.

**AI Output:**

See 23127152-hw6/test-cases/API1\_PoolA/generated.md rows TC-A1-001...016 (and robustness 017–018 drafted with domain misuse) — AI-assisted generation under Step A, then merged into final file.

**Human review:** Kept partitions; clarified empty search= expectation as “document actual”.

## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.1 Step B State

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:38 +07

**Prompt:**

Step B only: state transitions for GET /api/products. If N/A for read-only list, say so and propose compensating idempotent-read cases instead of inventing a fake state machine.

**AI Output:**

N/A for FR-10-style machine. Compensating cases TC-A1-019, TC-A1-020 (idempotent GET / search does not mutate). Recorded in generated.md.

**Human review:** Accepted N/A + two compensating cases.

## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.1 Step C Security

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:39 +07

### Prompt:

Step C only: map SEC-01...07 to concrete HTTP tests for public GET /api/products. Emphasize SEC-05 SQLi on search. Include payloads and expected secure behavior (parameterized query), not implementation bugs as expected.

### AI Output:

TC-A1-021...030 in generated.md (SQLi variants, public auth probes, LIKE %).

**Human review:** Expected results written against SEC-05 (no dump / no HTML 500), not against current buggy SUT.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.1 Step D Schema

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:39 +07

### Prompt:

Step D only: schema validation cases for list/search responses — Content-Type, array root, required fields/types, empty array, error must not be raw HTML.

### AI Output:

TC-A1-031...040 in generated.md.

**Human review:** Accepted; total 40 ≥ 35.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.2 Audit

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:40 +07

### Prompt:

Using skill api-test-audit: label every TC in API1\_PoolA/generated.md VALID/INVALID/INCOMPLETE with reasoning and corrections. Human owns final labels. Spec-first; do not treat buggy implementation as expected for SEC-05.

### AI Output:

See 23127152-hw6/test-cases/API1\_PoolA/audit.md — draft labels assisted by AI, confirmed by human (32 VALID / 3 INVALID / 5 INCOMPLETE).

**Human review:** Adjusted TC-A1-017 rewrite to DELETE without id; pinned empty-search behavior.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.3 Extend

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:41 +07

### Prompt:

Using skill api-test-extend: propose gaps AI missed for FR-05 after audit, focus security/encoding. I will keep ≥5 human-chosen rows only.

## AI Output:

Candidate gaps (encoding, Content-Type on error, row-count oracle, sequenced DROP). Human selected TC-A1-E01...E06 in extended.md.

**Human review:** Kept 6 extended TCs with taxonomy codes.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.4 Execute

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:40 +07

### Prompt:

Using skill api-test-execute: implement FR-05 Postman folder with assertions mapped to audited/extended TCs, ensure X-Student-Id: 23127152, run Newman + htmlextra against localhost:3000.

## AI Output:

Artifacts:

- postman/EShop-HW06.postman\_collection.json (20 FR-05 requests)
- postman/reports/newman-api1-fr05.html
- postman/screenshots/newman-api1-fr05-cli.txt (includes X-Student-Id=23127152)
- test-cases/API1\_PoolA/execution-notes.md

Result: 20 requests, 26 assertions, **4 failed** (SEC-05 bugs).

**Human review:** Failures accepted as genuine SEC-05 defects → bug reports.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 1.5 Bugs

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:42 +07

### Prompt:

Using skill bug-report + HW5 template: file BUG-PRODUCTS-001 (SQLi dump) and BUG-PRODUCTS-002 (HTML 500) with evidence; open GitHub Issues.

## AI Output:

Local MD: bug-reports/BUG-PRODUCTS-001.md, BUG-PRODUCTS-002.md

Issues: <https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/issues/300> · <https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/issues/301> (Anhnguyenk835)

**Human review:** Confirmed evidence paths and issue links.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 2.1 Generate FR-11

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:47 +07

### Prompt:

Using api-test-generate Steps A–D separately for FR-11: GET /api/orders/my-orders (JWT) and GET /api/orders/:id (no auth per selection). Cover domain, compensating state (status visibility), SEC-02 + IDOR, schema. ≥35 TCs → API2\_PoolB/generated.md.

## AI Output:

See 23127152-hw6/test-cases/API2\_PoolB/generated.md (40 TCs TC-B2-001...040).

**Human review:** Kept; IDOR expected = deny despite public detail surface.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 2.2 Audit

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:48 +07

### Prompt:

Using api-test-audit: label all FR-11 generated TCs VALID/INVALID/INCOMPLETE; human owns labels.

## AI Output:

API2\_PoolB/audit.md — 34 VALID / 2 INVALID / 4 INCOMPLETE.

**Human review:** Confirmed.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 2.3 Extend

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:48 +07

### Prompt:

Using api-test-extend: gaps for FR-11 IDOR/PII; human selects  $\geq 5$ .

## AI Output:

Candidates; human kept TC-B2-E01...E06 in extended.md.

**Human review:** Accepted 6 extended TCs.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 2.4–2.5 Execute + Bugs

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:50 +07

### Prompt:

Using api-test-execute + bug-report: implement FR-11 Postman folder (setup A/B orders + assertions), Newman run, file IDOR bug + GitHub Issue.

## AI Output:

- postman/reports/newman-api2-fr11.html — 26 requests, 35 assertions, 5 failed (all IDOR)
- bug-reports/BUG-ORDERS-001.md
- Issue: <https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/issues/302> (Anhnguyenk835)

**Human review:** Confirmed PII leak Addr B Secret in unauth response.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 3.1–3.3 Generate/Audit/Extend FR-15

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:52 +07

**Prompt:**

Using `api-test-generate A-D` then `api-test-audit` then `api-test-extend` for FR-15 POST/PUT/DELETE /api/products (admin). Spec requires JWT+admin; emphasize SEC-02/SEC-03.

**AI Output:**

- `API3_PoolC/generated.md` (40 TCs)
- `audit.md` (33 VALID / 3 INVALID / 4 INCOMPLETE)
- `extended.md` (TC-C3-E01...E06)

**Human review:** Security expected = 401/403 per spec, not current unauthenticated success.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 3.4–3.5 Execute + Bugs

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 21:54 +07

**Prompt:**

Using `api-test-execute + bug-report`: Newman FR-15 suite; file SEC-02/03 + validation + DELETE semantics bugs and GitHub Issues.

**AI Output:**

- `newman-api3-fr15.html` — 22 req, 25 assertions, 12 failed
- Issues: #303 (auth), #304 (validation), #305 (DELETE 200 on missing) — Anhnguyenk835

**Human review:** Confirmed unauth create returns 200 with new id.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 4 Postman/Newman final

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 22:04 +07

**Prompt:**

Using `api-test-execute`: finalize Phase 4 — data-driven CSV run, consolidated `newman-report.html`, header evidence, list Postman features, update `scripts/run-newman.sh`.

**AI Output:**

- Data-driven: 8 iterations, 32 assertions, 0 failed → `newman-report-data-driven.html`
- Full collection → `newman-report.html` (failures = known Phase1–3 bugs)
- `postman/PHASE4_NOTES.md`, mock example, README feature table updated

**Human review:** Accepted; Monitors deferred to CI/CD Phase 5.

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## Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 5 CI/CD

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 22:16 +07

**Prompt:**

Using `api-test-cicd`: add GitHub Actions Newman workflow, CI Smoke folder, push pass + intentional fail sample runs, document with screenshots/links.

#### AI Output:

- Workflow: `.github/workflows/hw06-api-tests.yml`
- Pass: commit `8b44b66` → Actions run `32269053058` (success)
- Fail: commit `2fe51ee` → Actions run `32269286455` (failure, CI-01 expect 201)
- Reports: `report/CI_CD_Report.md`, `cicd/sample-commits.md`, screenshots

**Human review:** Confirmed live Actions conclusions; restored `ciExpectStatus=200`.

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### Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 6 Agent Skill design

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 23:50 +07

#### Prompt:

Using `api-test-generator-design`: refine `agent-skill/pseudocode.md` to match HW06 pipeline (parser → domain/state/security/schema → human audit → extend → Postman/Newman). Keep diagram topology from hand-authored `diagram.mmd` / SVG.

#### AI Output:

Updated `23127152-hw6/agent-skill/pseudocode.md` (components table + extended pseudocode). Diagram PNG from student-authored SVG/Mermaid topology (`diagram.png`).

**Human review:** Confirmed human audit gate and  $\geq 5$  extend hook remain mandatory on diagram.

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### Session — Cursor (Grok 4.5) — 2026-08-19 — Phase 7 Reports + Critique

**AI Tool:** Cursor (Cursor Grok 4.5)

**Date/Time:** 2026-08-19 23:52 +07

#### Prompt:

Using `hw06-main-report` + `ai-critique`: fill main report, AI Critique 200–300 words grounded in extended TC oracles / bugs #300–#305, update README self-assessment, refresh CI/CD screenshot paths, export `git-commit-log`.

#### AI Output:

- `report/23127152_HW06_Report.md`
- `ai-audit/AI_Critique.md` (268 words)
- README self-assessment + evidence links
- `git-commit-log.txt`

**Human review:** Counts cross-checked with Newman notes and Issues #300–#305 (Anhnguyenk835).